
Knowledge Discovery & Data Mining

— Classifier Evaluation, Model Selection —

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Outline: Model Evaluation and Selection

- Evaluation metrics: How can we measure accuracy? Other metrics to consider?
- Use **validation set** of class-labeled tuples instead of training set when assessing accuracy
- Methods for estimating a classifier's accuracy:
 - Holdout method, random subsampling
 - Cross-validation
 - Bootstrap
- Comparing classifiers:
 - Confidence intervals
 - Cost-benefit analysis and ROC Curves

Classifier Evaluation Metrics: Confusion Matrix

Confusion Matrix:

Actual class\Predicted class	C_1	$\neg C_1$
C_1	True Positives (TP)	False Negatives (FN)
$\neg C_1$	False Positives (FP)	True Negatives (TN)

Example of Confusion Matrix:

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
Total	7366	2634	10000

- Given m classes, an entry, $CM_{i,j}$ in a **confusion matrix** indicates # of tuples in class i that were labeled by the classifier as class j
- May have extra rows/columns to provide totals

Accuracy, Error Rate, Sensitivity and Specificity

A\P	C	¬C	
C	TP	FN	P
¬C	FP	TN	N
	P'	N'	All

- **Classifier Accuracy**, or recognition rate: percentage of test set tuples that are correctly classified

$$\text{Accuracy} = (TP + TN)/All$$

- **Error rate(misclassification rate)**: $1 - \text{accuracy}$, or
 $\text{Error rate} = (FP + FN)/All$

- **Class Imbalance Problem:**

- One class may be *rare*, e.g. fraud, or HIV-positive
- Significant *majority of the negative class* and minority of the positive class

- **Sensitivity**: True Positive recognition rate (the proportion of positive tuples that are correctly identified)

- **Sensitivity** = TP/P

- **Specificity**: True Negative recognition rate (the proportion of negative tuples that are correctly identified)

- **Specificity** = TN/N

Precision and Recall, and F-measures

- **Precision:** exactness – what % of tuples that the classifier labeled as positive are actually positive

$$precision = \frac{TP}{TP + FP}$$

- **Recall:** completeness – what % of positive tuples did the classifier label as positive?

- Perfect score is 1.0

- Inverse relationship between precision & recall

$$recall = \frac{TP}{TP + FN} = \frac{TP}{P}$$

- **F measure (F_1 or F-score):** harmonic mean of precision and recall,

$$F = \frac{2 \times precision \times recall}{precision + recall}$$

- F_β : weighted measure of precision and recall

- assigns β times as much weight to recall as to precision

$$F_\beta = \frac{(1 + \beta^2) \times precision \times recall}{\beta^2 \times precision + recall}$$

Classifier Evaluation Metrics: Example

Actual Class\Predicted class	cancer = yes	cancer = no	Total	Recognition(%)
cancer = yes	90	210	300	30.00 (<i>sensitivity</i>)
cancer = no	140	9560	9700	98.56 (<i>specificity</i>)
Total	230	9770	10000	96.40 (<i>accuracy</i>)

Measure	Formula
accuracy, recognition rate	$\frac{TP + TN}{P + N}$
error rate, misclassification rate	$\frac{FP + FN}{P + N}$
sensitivity, true positive rate, recall	$\frac{TP}{P}$
specificity, true negative rate	$\frac{TN}{N}$
precision	$\frac{TP}{TP + FP}$
F , F_1 , F -score, harmonic mean of precision and recall	$\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

		Predicted class		
		<i>yes</i>	<i>no</i>	Total
Actual class	<i>yes</i>	TP	FN	P
	<i>no</i>	FP	TN	N
	Total	P'	N'	$P + N$

- $Precision = 90/230 = 39.13\%$
- $Recall = 90/300 = 30.00\%$



Issues Affecting Model Selection

- **Speed**
 - time to construct the model (training time)
 - time to use the model (classification/prediction time)
- **Robustness**: handling noise and missing values
- **Scalability**: is typically assessed with a series of data sets of increasing size.
- **Interpretability**
 - understanding and insight provided by the model
- Other measures, e.g., goodness of rules, such as decision tree size or compactness of classification rules

Evaluation: Holdout & Cross-Validation Methods

■ Holdout method

- Given data is randomly partitioned into two independent sets
 - Training set (e.g., 2/3) for model construction
 - Test set (e.g., 1/3) for accuracy estimation
- Random sampling: a variation of holdout
 - Repeat holdout k times, accuracy = avg. of the accuracies obtained

■ Cross-validation (k -fold, where $k = 10$ is most popular)

- Randomly partition the data into k *mutually exclusive* subsets, each approximately equal size
- At i -th iteration, use D_i as test set and others as training set
- Leave-one-out: k folds where $k = \#$ of tuples, for small sized data
- *Stratified cross-validation*: folds are stratified so that class dist. in each fold is approx. the same as that in the initial data

Evaluation: Bootstrap

- **Bootstrap**

- Works well with small data sets
- Samples the given training tuples uniformly *with replacement*
 - i.e., each time a tuple is selected, it is equally likely to be selected again and re-added to the training set

- Several bootstrap methods, and a common one is **.632 bootstrap**

- A data set with d tuples is sampled d times, with replacement, resulting in a training set of d samples. The data tuples that did not make it into the training set end up forming the test set. About 63.2% of the original data end up in the bootstrap, and the remaining 36.8% form the test set (since $(1 - 1/d)^d \approx e^{-1} = 0.368$)
- Repeat the sampling procedure k times, overall accuracy of the model:

$$Acc(M) = \frac{1}{k} \sum_{i=1}^k (0.632 \times Acc(M_i)_{test_set} + 0.368 \times Acc(M_i)_{train_set})$$

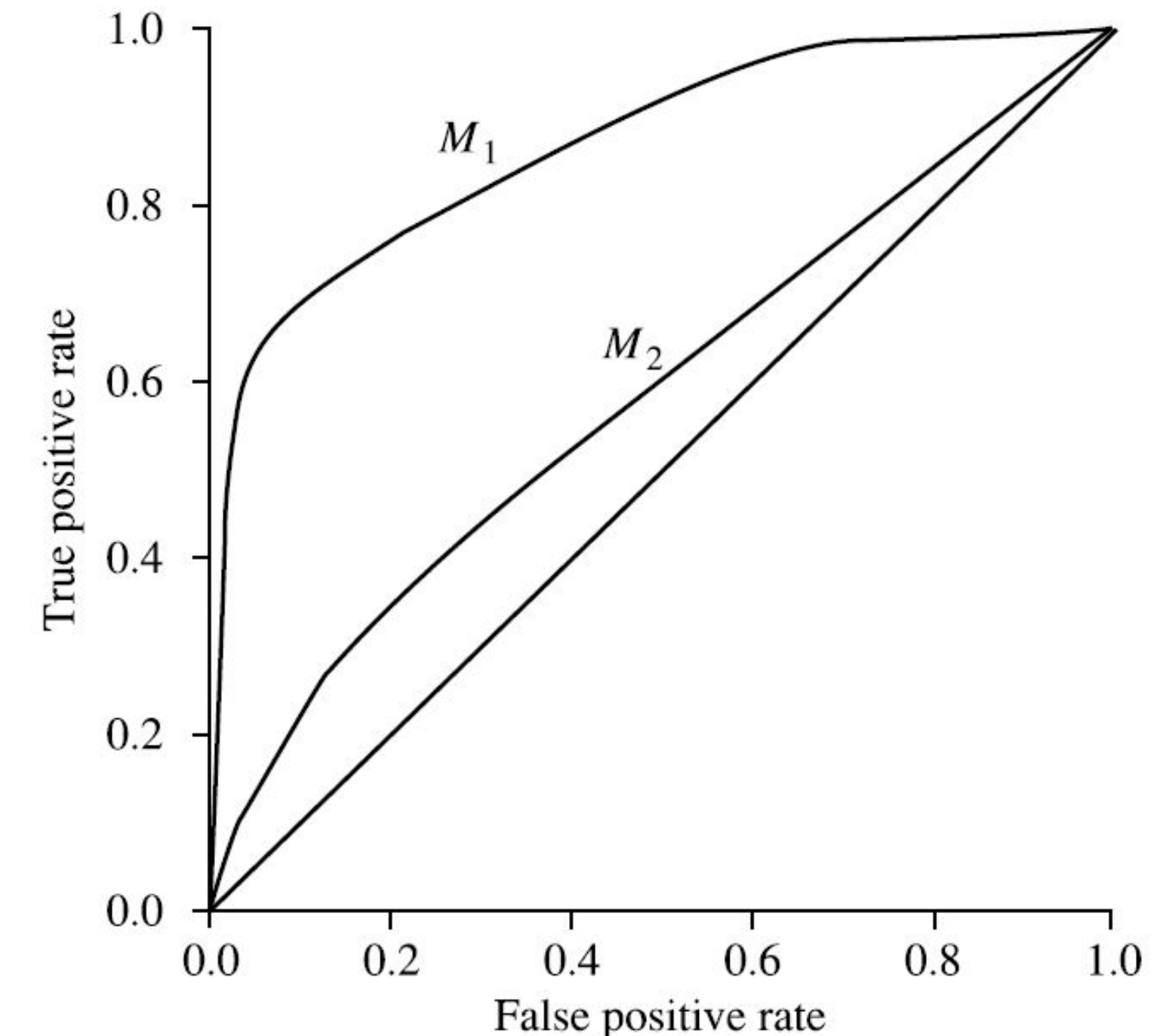
where $Acc(M_i)_{test_set}$ is the accuracy of the model obtained with bootstrap sample i when it is applied to test set i .

$Acc(M_i)_{train_set}$ is the accuracy of the model obtained with bootstrap sample i when it is applied to the original set of data tuples.

The .632 bootstrap

Model Selection: ROC Curves

- **ROC** (Receiver Operating Characteristics) curves: for visual comparison of classification models
- Originated from signal detection theory
- Shows the trade-off between the true positive rate and the false positive rate
- The area under the ROC curve is a measure of the accuracy of the model
- Rank the test tuples in decreasing order: the one that is most likely to belong to the positive class appears at the top of the list.
- The closer to the diagonal line (i.e., the closer the area is to 0.5), the less accurate is the model



- Vertical axis represents the **true positive rate**
- Horizontal axis rep. the **false positive rate**
- The plot also shows a diagonal line
- A model with perfect accuracy will have an area of 1.0

Model Selection: ROC Curves

Example. Plotting a ROC curve. The following figure shows the probability value (column 3) returned by a probabilistic classifier for each of the 10 tuples in a test set, sorted in the decreasing probability order.

Tuple #	Class	Prob.	<i>TP</i>	<i>FP</i>	<i>TN</i>	<i>FN</i>	<i>TPR</i>	<i>FPR</i>
1	<i>P</i>	0.90						
2	<i>P</i>	0.80						
3	<i>N</i>	0.70						
4	<i>P</i>	0.60						
5	<i>P</i>	0.55						
6	<i>N</i>	0.54						
7	<i>N</i>	0.53						
8	<i>N</i>	0.51						
9	<i>P</i>	0.50						
10	<i>N</i>	0.40						

Predicted class			
Actual class	<i>yes</i>	<i>no</i>	Total
	<i>TP</i>	<i>FN</i>	<i>P</i>
	<i>FP</i>	<i>TN</i>	<i>N</i>
Total	<i>P'</i>	<i>N'</i>	<i>P + N</i>

$TPR = TP / (TP + FN)$

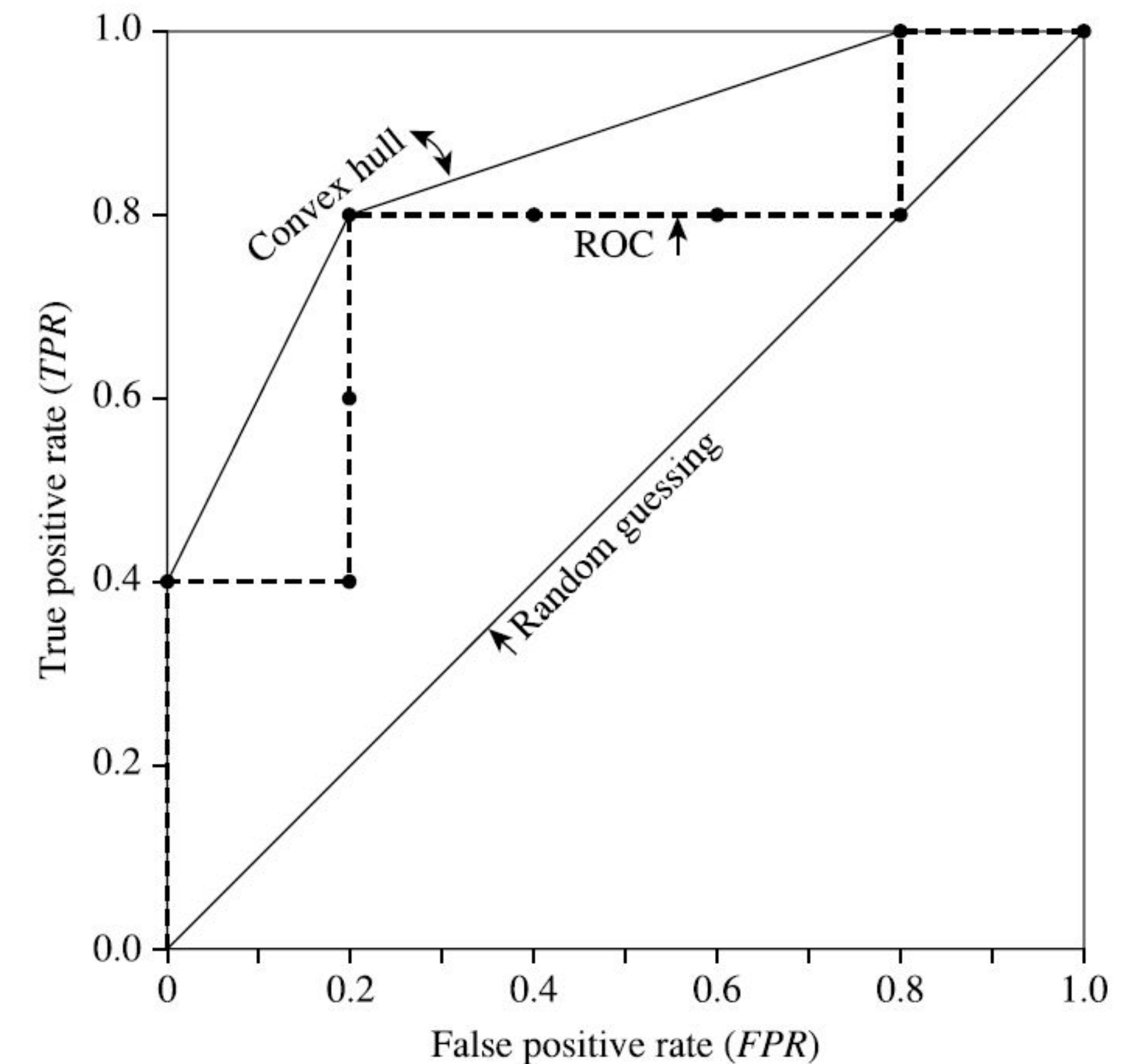
$FPR = FP / (FP + TN)$



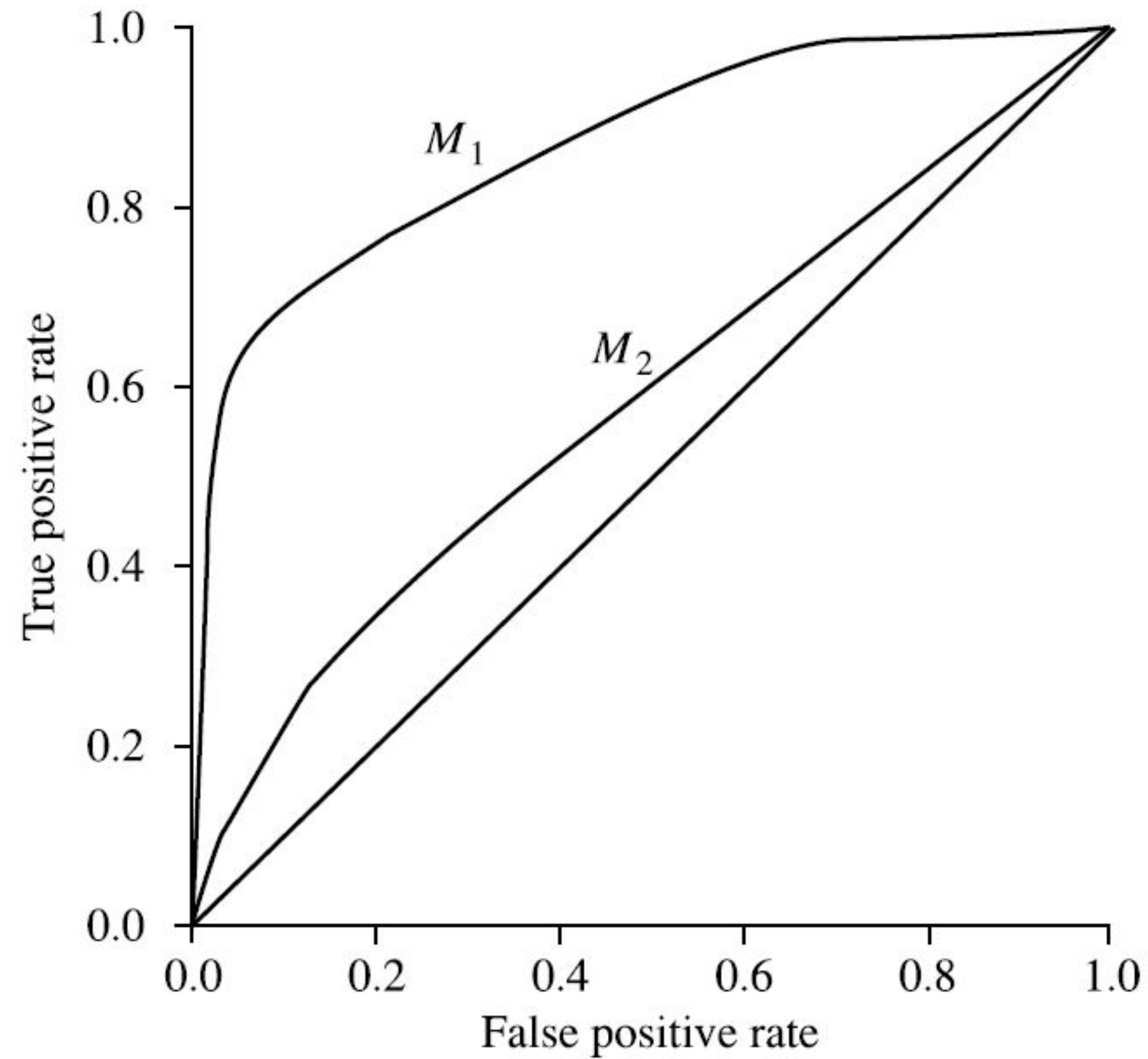
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1	<i>P</i>	0.90	1	0	5	4	0.2	0
2	<i>P</i>	0.80	2	0	5	3	0.4	0
3	<i>N</i>	0.70	2	1	4	3	0.4	0.2
4	<i>P</i>	0.60	3	1	4	2	0.6	0.2
5	<i>P</i>	0.55	4	1	4	1	0.8	0.2
6	<i>N</i>	0.54	4	2	3	1	0.8	0.4
7	<i>N</i>	0.53	4	3	2	1	0.8	0.6
8	<i>N</i>	0.51	4	4	1	1	0.8	0.8
9	<i>P</i>	0.50	5	4	1	0	1.0	0.8
10	<i>N</i>	0.40	5	5	0	0	1.0	1.0



Model Selection: ROC Curves



Summary

- Evaluation metrics
 - Confusion Matrix, Accuracy, Error Rate,
 - Sensitivity and Specificity
 - Precision and Recall, and F-measures
 - Issues Affecting Model Selection
- Methods for estimating a classifier's accuracy:
 - Holdout method, random subsampling
 - Cross-validation
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- Comparing classifiers:
 - Cost-benefit analysis and ROC Curves